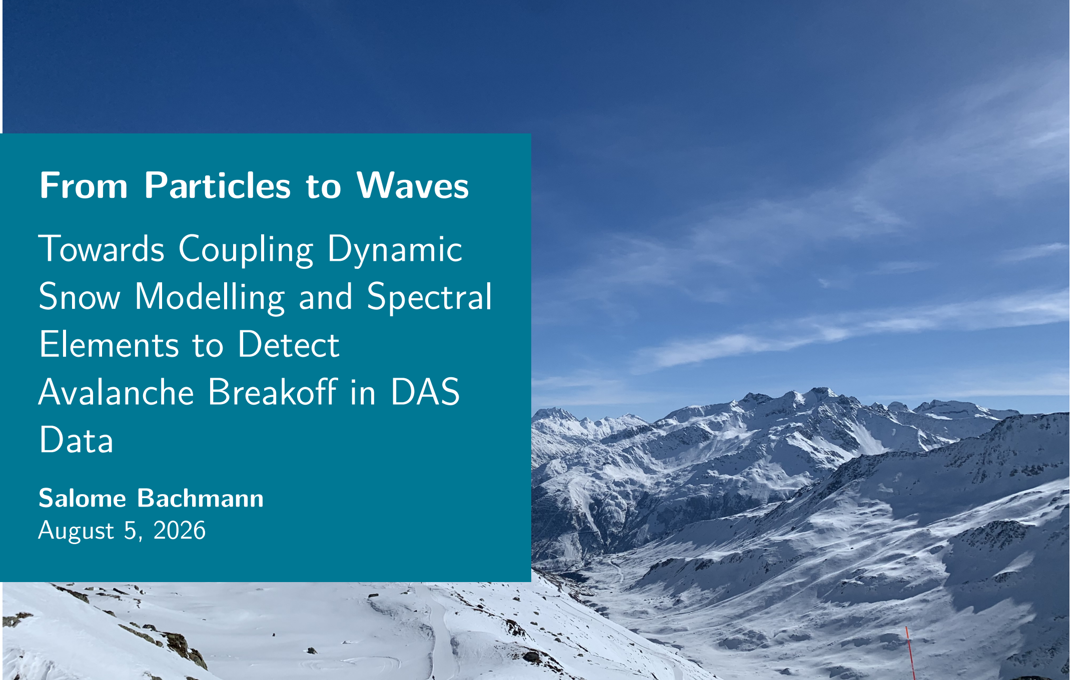

From Particles to Waves

Towards Coupling Dynamic Snow Modelling and Spectral Elements to Detect Avalanche Breakoff in DAS Data

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August 5, 2026

Overview

- Created 3 different models and 2 scenarios for crack propagation in snow using SALVUS
- Two pure spectral element (SE) models, one combined with data from material point method (MPM) simulations
- **Goal**: Couple dynamic snow modelling with SE to model crack propagation in snow

Why model crack propagation in snow? - Relevance

- 22 fatalities by avalanche in Switzerland every year (SLF, 2025)
- Avalanches are danger for human life but also infrastructure (Schweizer et al., 2003)
- Understanding processes associated with avalanches important
- Avalanche release because of loading, leading to crack propagation in a weal layer (Gaume et al., 2017)

Relevance

- Crack propagation at different speeds: sub-Rayleigh (below vs of material) and Supershear (above vs) (Trottet et al., 2022), transition between two regimes possible (Bergfeld et al., 2025)
- Fast crack propagation $\rightarrow$critical crack length for release (Gaume et al., 2015) is reached faster
- Distributed Acoustic Sensing (DAS) already used to monitor avalanches in Switzerland (Edme et al., 2023; Paitz et al., 2023) and Norway (Turquet et al., 2024)
- No knowledge if crack propagation associated with breakoff can be seen in DAS data

Research Questions

- What FE framework is required to to simulate DAS measurements of crack propagation in snow, and what are the key assumptions and limitations?
- What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?
- To what extent do the results change if source wavelets from a MPM simulation of a propagation saw test (PST) are injected?
- How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?

Simple Model I

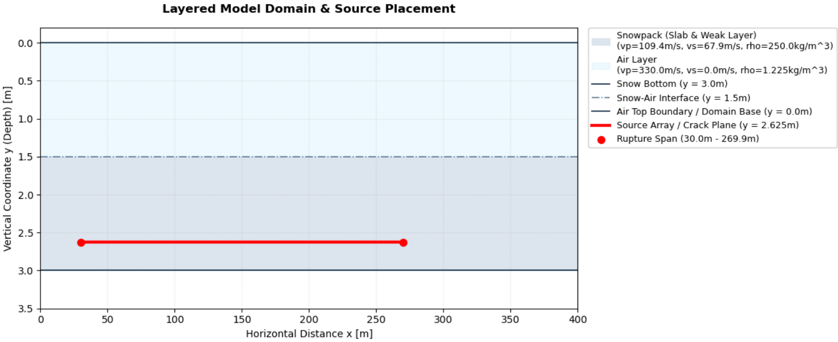

Figure: Two-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red.

Simple Model II

- Set up to test custom STF in SALVUS, see physical effects on output
- Sources spaced over 240m in 10cm intervals
- 20Hz central frequency Ricker wavelet
- Source Wavelet in xx (1), xy (1) and yy (-1) direction
- Time delay of wavelet calculated based on propagation speed: sub-Rayleigh speed is $0.6v_s$, Supershear $1.6v_s$ (Trottet et al., 2022)

MPM-Guided Model I

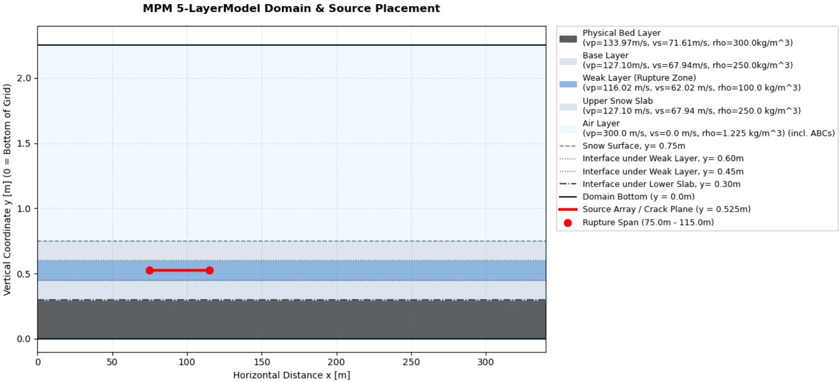

Figure: Five-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red.

MPM-Guided Model II

- Source-time functions obtained from MPM-simulation of PST by Trottet et al., 2022
- Source Wavelet in all three directions, but irregular and different for every source point

Results

- SALVUS Output: Displacement field u
- $\frac{\partial u_i}{\partial t}$ $\rightarrow$velocity in x and y
- $\frac{\partial u_i}{\partial j}$ $\rightarrow$strain in x and y ϵ
- $\frac{\partial^2 u_i}{\partial t \partial t}$ $\rightarrow$strain rate $\dot{\epsilon}$
- Simulation od DAS data from ϵ, $\dot{\epsilon}$ using Pyber Gauge length class by **noeFrontiersFullwaveformSeismology2025**, gauge length acoording to Turquet et al., 2024

Simple Sub-Rayleigh: Strain and Strain Rate I

Simple Supershear: Strain and Strain Rate I

MPM-Guided: Strain and Strain Rate I

Summary of Results & Discussion I

Implications for DAS field data I

Outlook

Answer to Research Questions - Conclusions

Questions?

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MPM STF Workflow

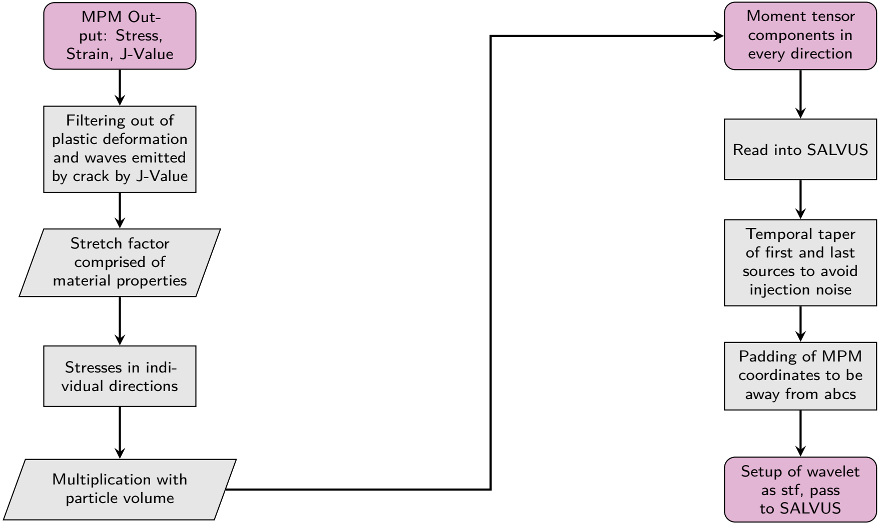

Figure: Workflow for conversion of MPM data and input as SALVUS STF.

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